Javier Vela Tambo

javier.vela00@gmail.com | +1 (508) 579-9416 | Worcester, MA
javiervela | javiervelatambo | javiervelatambo.com

Computer Science PhD Candidate researching **Distributed Quantum Computing (DQC)** compilation, circuit partitioning, and simulation. Looking for an internship to connect my PhD research with industry, using my **Data Science**, **Machine Learning** and **Cloud** skills to solve practical quantum challenges while bringing real-world insights back to my research.

Work Experience

Research Assistant | Quantum Computing Worcester Polytechnic Institute, MA, USA Aug 2025 – Present

- Research compilation, circuit partitioning algorithms, and simulation for Distributed Quantum Computing (DQC).
- Benchmarked 6 partitioners across 1,000+ circuits and 4 network topologies for structural and communication trade-offs.
- Authored educational materials for a Quantum Computing course and co-mentored an undergraduate thesis.
- Published and presented research at the IEEE QCE 2026 DQC-SI Workshop.

Data Scientist | NLP Pyrenean Institute of Ecology (IPE-CSIC), Spain Jun 2024 – Aug 2025

- Researched and benchmarked open-source LLM capabilities for extracting structured data from unstructured text.
- Formulated a novel methodology using autoregressive LLMs to extract structured meteorological impact metrics.
- Engineered end-to-end data pipelines to download, process, and analyze millions of articles from Spanish news corpora.

DevOps Engineer | CI/CD Automation Developer Ericsson, Spain Jul 2022 – Jun 2024

- Architected CI/CD pipelines and Kubernetes deployments, reducing deployment overhead across 10+ development teams.
- Provisioned and maintained multi-cloud infrastructure across AWS and Azure using Terraform.
- Designed and deployed an observability platform and dashboards using Prometheus and Grafana.

Education

PhD in Computer Science	Worcester Polytechnic Institute, USA		Aug 2025 – Present
MS in AI Research	Menéndez Pelayo International University, Spain	GPA: 9/10	Oct 2024 – Sep 2025
BS in Computer Engineering	University of Zaragoza, Spain	GPA: 8.23/10	Sep 2018 – Jun 2023

Selected Publications

Javier Vela-Tambo, Davud Azizov, Tian Guo (Aug. 2026). *Towards Reproducible Evaluation of Distributed Quantum Circuit Partitioning Algorithms*. DOI: 10.48550/arXiv.2608.27099. arXiv: 2608.27099 [quant-ph]. ***(In Proc. IEEE QCE 2026)***.

Javier Vela-Tambo, Jorge Gracia, Fernando Dominguez-Castro (Dec. 2025). *CienaLLM: Generative Climate-Impact Extraction from News Articles with Autoregressive LLMs*. DOI: 10.48550/arXiv.2512.19305. arXiv: 2512.19305 [cs]. ***(Submitted to TACL)***.

Javier Vela, Unai Arronategui, José Ángel Bañares, José Manuel Colom (2025). "Distributed Simulation with Efficient Fault Tolerance". In: *Economics of Grids, Clouds, Systems, and Services*. Vol. 15358. Springer Nature Switzerland, pp. 261–274. DOI: 10.1007/978-3-031-81226-2_23. ***(Undergraduate Thesis awarded with honors)***.

Skills

Programming	Python, Rust, C/C++, Bash	**Visualization**	Matplotlib, Seaborn
Quantum	Qiskit, PennyLane, DQC	**Observability**	Prometheus, Grafana
Machine Learning	PyTorch, TensorFlow, scikit-learn	**HPC**	Slurm, OpenMPI, OpenMP
Data Science	Pandas, NumPy, SciPy	**CI/CD**	GitHub Actions, GitLab CI, Jenkins
Cloud & Infra	Kubernetes, Docker, AWS, Terraform	**Databases**	SQL, MongoDB
Systems & Dev	Linux, Git, GitHub, CLI, KVM/QEMU	**Languages**	English, Spanish

Activities

IBM Qiskit Advocate Since Nov 2025
Qiskit Global Summer School 2026: Excellence Badge Achieved Jul 2026